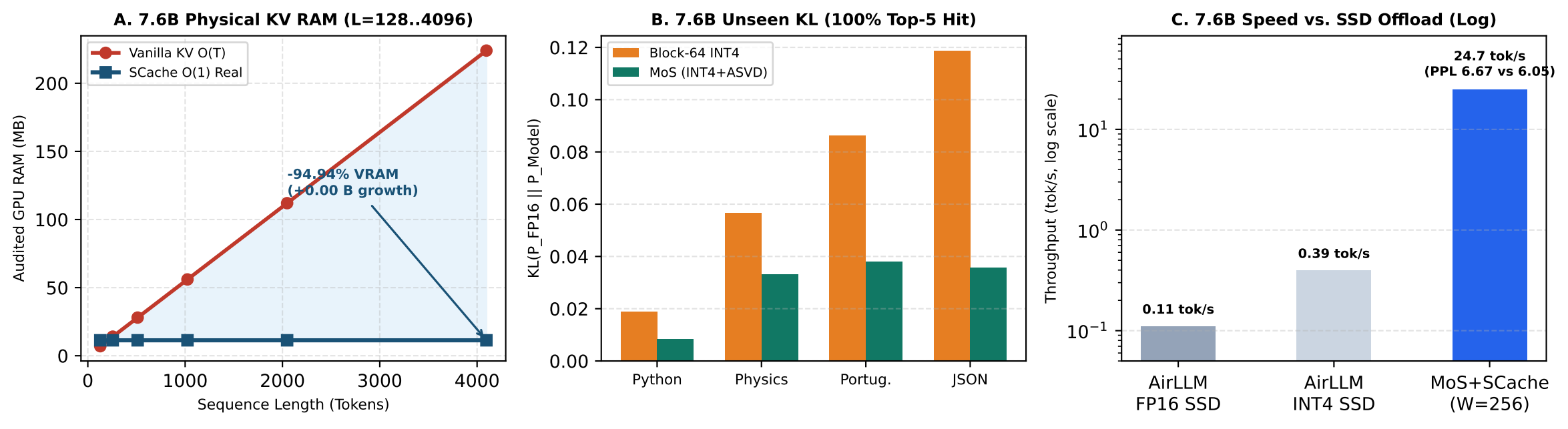
A. 7.6B Physical KV RAM (L=128..4096)
Vanilla KV O(T)
SCache O(1) Real
-94.94% VRAM
(+0.00 B growth)
Audited GPU RAM (MB)
Sequence Length (Tokens)
0
50
100
150
200
0
1000
2000
3000
4000
B. 7.6B Unseen KL (100% Top-5 Hit)
Block-64 INT4
MoS (INT4+ASVD)
KL(P_FP16 || P_Model)
0.00
0.02
0.04
0.06
0.08
0.10
0.12
Python
Physics
Portug.
JSON
C. 7.6B Speed vs. SSD Offload (Log)
Throughput (tok/s, log scale)
0.11 tok/s
0.39 tok/s
24.7 tok/s
(PPL 6.67 vs 6.05)
AirLLM
FP16 SSD
AirLLM
INT4 SSD
MoS+SCache
(W=256)